



TULSIRAMJI GAIKWAD-PATIL College of Engineering and Technology

Wardha Road, Nagpur - 441108

Accredited with NAAC A+ Grade

Approved by AICTE, New Delhi, Govt. of Maharashtra



(An Autonomous Institute Affiliated to RTM Nagpur University with NAAC A+ Grade)

Department of Information Technology

Session 2023-24(Even Semester)

| VISION                                                                                       | MISSION                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| To emerge as a learning hub and center of excellence in the domain of Information Technology | [M1] To impart quality technical education through effective teaching learning process.<br>[M2] To provide a platform to address societal issues as well as challenges faced by IT industries.<br>[M3] To foster a culture of research and impart innovative and entrepreneurial skills in the field of IT.<br>[M4] To ensure overall development of students and staff by inculcating knowledge and professional ethics as a part of lifelong learning. |

### Class Test 2

Semester: VI

Programme: B. Tech Information Technology

Course: Artificial Intelligence and Machine Learning

Time: 1 hours 60min

Max Marks: 30 Marks

#### Instructions to candidates:

1. All questions are compulsory in Group A
2. Solve any two sub-question out of three in Group B
3. Assume suitable data wherever necessary
4. All question carries marks as indicated

| BIT33602 | Course Outcome                                                                                      |
|----------|-----------------------------------------------------------------------------------------------------|
| CO1      | Identify knowledge representation, problem solving and learning methods of AI.                      |
| CO2      | Discuss knowledge and reasoning and its representation in AI.                                       |
| CO3      | Describe artificial intelligence, machine learning and deep learning                                |
| CO4      | Analyze supervised learning algorithms and decision trees.                                          |
| CO5      | Evaluate machine learning models using cross-validation and sampling techniques like bootstrapping. |

| Group A |                                               |     |     |       |
|---------|-----------------------------------------------|-----|-----|-------|
| Que. 1  | Answer the following Question                 | CO  | BTL | Marks |
| a.      | Define <b>Overfitting</b> .                   | CO3 | 2   | 2     |
| b.      | What is <b>RNN</b> ?                          | CO4 | 2   | 2     |
| c.      | What is <b>Bootstrapping</b> ?                | CO5 | 3   | 2     |
| Group B |                                               |     |     |       |
| Que. 2  | Answer the following Question (Solve any Two) | CO  | BTL | Marks |
| a.      | Explain <b>Neural Network basics</b> .        | CO3 | 2   | 4     |
| b.      | Write note on <b>Clustering problems</b> .    | CO3 | 2   | 4     |
| c.      | Explain <b>Deep Learning concept</b> .        | CO3 | 2   | 4     |

|        |                                                       |     |     |       |
|--------|-------------------------------------------------------|-----|-----|-------|
| Que 3. | Answer the following Question (Solve any Two)         | CO  | BTL | Marks |
| a.     | Explain <b>Difference between CNN and RNN</b> .       | CO4 | 4   | 4     |
| b.     | Explain <b>Support Vector Machine (SVM)</b> .         | CO4 | 3   | 4     |
| c.     | Explain <b>Regression Decision Trees</b> .            | CO4 | 3   | 4     |
| Que 4. | Answer the following Question (Solve any Two)         | CO  | BTL | Marks |
| a.     | Explain <b>LDA (Linear Discriminant Analysis)</b> .   | CO5 | 2   | 4     |
| b.     | What is <b>K-fold Cross Validation</b> ?              | CO5 | 3   | 4     |
| c.     | Explain <b>Independent Component Analysis (ICA)</b> . | CO5 | 3   | 4     |

**Bloom's Taxonomy Level (BTL)-** Remember , Understand , Apply, Analyze , Evaluate , Create